

Commerce XI Maths App

Exercise 3.2

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 3 - Complex Numbers | Total questions: 17

Question 1 - Q1(i)

Find the square root of complex number, $-8 - 6i$

Options

- A. $\pm(1 - 3i)$
- B. $\pm(1 + 3i)$
- C. $\pm(3 - i)$
- D. $\pm(2 - 2i)$

Correct answer: $\pm(1 - 3i)$

Explanation and steps

Let $a + ib$ be a square root. Expand $(a + ib)^2 = a^2 - b^2 + 2abi$, equate real and imaginary parts, and include both signs. The roots are $\pm(1 - 3i)$.

Question 2 - Q1(ii)

Find the square root of complex number, $7 + 24i$

Options

- A. $\pm(4 + 3i)$
- B. $\pm(4 - 3i)$
- C. $\pm(3 + 4i)$
- D. $\pm(7 + 12i)$

Correct answer: $\pm(4 + 3i)$

Explanation and steps

Let $a + ib$ be a square root. Expand $(a + ib)^2 = a^2 - b^2 + 2abi$, equate real and imaginary parts, and include both signs. The roots are $\pm(4 + 3i)$.

Question 3 - Q1(iii)

Find the square root of complex number, $1 + 4\sqrt{3}i$

Options

- A. $\pm(2 + \sqrt{3})i$
- B. $\pm(2 - \sqrt{3})i$
- C. $\pm(\sqrt{3} + 2i)$
- D. $\pm(1 + 2\sqrt{3})i$

Correct answer: $\pm(2 + \sqrt{3})i$

Explanation and steps

Let $a + ib$ be a square root. Expand $(a + ib)^2 = a^2 - b^2 + 2abi$, equate real and imaginary parts, and include both signs. The roots are $\pm(2 + \sqrt{3})i$.

Question 4 - Q1(iv)

Find the square root of complex number, $3 + 2\sqrt{10}i$

Options

A. $\pm(\sqrt{5} + \sqrt{2}i)$

B. $\pm(\sqrt{5} - \sqrt{2}i)$

C. $\pm(\sqrt{10} + i)$

D. $\pm(\sqrt{2} + \sqrt{5}i)$

Correct answer: $\pm(\sqrt{5} + \sqrt{2}i)$

Explanation and steps

Let $a + ib$ be a square root. Expand $(a + ib)^2 = a^2 - b^2 + 2abi$, equate real and imaginary parts, and include both signs. The roots are $\pm(\sqrt{5} + \sqrt{2}i)$.

Question 5 - Q1(v)

Find the square root of complex number, $2(1 - \sqrt{3}i)$

Options

A. $\pm(\sqrt{3} - i)$

B. $\pm(\sqrt{3} + i)$

C. $\pm(1 - \sqrt{3}i)$

D. $\pm(2 - \sqrt{3}i)$

Correct answer: $\pm(\sqrt{3} - i)$

Explanation and steps

Let $a + ib$ be a square root. Expand $(a + ib)^2 = a^2 - b^2 + 2abi$, equate real and imaginary parts, and include both signs. The roots are $\pm(\sqrt{3} - i)$.

Question 6 - Q2(i)

Solve quadratic equation, $8x^2 + 2x + 1 = 0$

Options

A. $x = (-1 \pm \sqrt{7}i)/(8)$

B. $x = (1 \pm \sqrt{7}i)/(8)$

C. $x = (-1 \pm 7i)/(8)$

D. $x = (-2 \pm \sqrt{7}i)/(8)$

Correct answer: $x = (-1 \pm \sqrt{7}i)/(8)$

Explanation and steps

Apply the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = (-1 \pm \sqrt{7}i)/(8)$.

Question 7 - Q2(ii)

Solve quadratic equation, $2x^2 - \sqrt{3}x + 1 = 0$

Options

- A. $x = (\sqrt{3} \pm \sqrt{5} i)/(4)$
- B. $x = (-\sqrt{3} \pm \sqrt{5} i)/(4)$
- C. $x = (\sqrt{3} \pm \sqrt{5})/(4)$
- D. $x = (\sqrt{3} \pm i)/(2)$

Correct answer: $x = (\sqrt{3} \pm \sqrt{5} i)/(4)$

Explanation and steps

Apply the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = (\sqrt{3} \pm \sqrt{5} i)/(4)$.

Question 8 - Q2(iii)

Solve quadratic equation, $3x^2 - 7x + 5 = 0$

Options

- A. $x = (7 \pm \sqrt{11} i)/(6)$
- B. $x = (-7 \pm \sqrt{11} i)/(6)$
- C. $x = (7 \pm 11i)/(6)$
- D. $x = (7 \pm \sqrt{11})/(6)$

Correct answer: $x = (7 \pm \sqrt{11} i)/(6)$

Explanation and steps

Apply the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = (7 \pm \sqrt{11} i)/(6)$.

Question 9 - Q2(iv)

Solve quadratic equation, $x^2 - 4x + 13 = 0$

Options

- A. $x = 2 \pm 3i$
- B. $x = -2 \pm 3i$
- C. $x = 4 \pm 13i$
- D. $x = 2 \pm i$

Correct answer: $x = 2 \pm 3i$

Explanation and steps

Apply the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = 2 \pm 3i$.

Question 10 - Q3(i)

Solve quadratic equation, $x^2 + 3ix + 10 = 0$

Options

- A. $x = 2i$ or $-5i$
- B. $x = -2i$ or $5i$
- C. $x = 2$ or -5
- D. $x = 5i$ or $-2i$

Correct answer: $x = 2i$ or $-5i$

Explanation and steps

Apply the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = 2i$ or $-5i$.

Question 11 - Q3(ii)

Solve quadratic equation, $2x^2 + 3ix + 2 = 0$

Options

- A. $x = (i)/(2)$ or $-2i$
- B. $x = (-i)/(2)$ or $2i$
- C. $x = i$ or $-i$
- D. $x = (1)/(2)$ or -2

Correct answer: $x = (i)/(2)$ or $-2i$

Explanation and steps

Apply the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = (i)/(2)$ or $-2i$.

Question 12 - Q3(iii)

Solve quadratic equation, $x^2 + 4ix - 4 = 0$

Options

- A. $x = -2i, -2i$
- B. $x = 2i, 2i$
- C. $x = -4i$ or i
- D. $x = \pm 2i$

Correct answer: $x = -2i, -2i$

Explanation and steps

Apply the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = -2i, -2i$.

Question 13 - Q3(iv)

Solve quadratic equation, $ix^2 - 4x - 4i = 0$

Options

- A. $x = -2i, -2i$
- B. $x = 2i, 2i$
- C. $x = \pm 2i$
- D. $x = -4i$ or 0

Correct answer: $x = -2i, -2i$

Explanation and steps

Apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = -2i, -2i$.

Question 14 - Q4(i)

Solve quadratic equation, $x^2 - (2 + i)x - (1 - 7i) = 0$

Options

- A. $x = 3 - i$ or $-1 + 2i$
- B. $x = 3 + i$ or $-1 - 2i$
- C. $x = 2 + i$ or $-1 + 7i$
- D. $x = 1 - i$ or $-3 + 2i$

Correct answer: $x = 3 - i$ or $-1 + 2i$

Explanation and steps

Apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = 3 - i$ or $-1 + 2i$.

Question 15 - Q4(ii)

Solve quadratic equation, $x^2 - (3\sqrt{2} + 2i)x + 6\sqrt{2}i = 0$

Options

- A. $x = 3\sqrt{2}$ or $2i$
- B. $x = -3\sqrt{2}$ or $-2i$
- C. $x = \sqrt{2}$ or $6i$
- D. $x = 3\sqrt{2}i$ or 2

Correct answer: $x = 3\sqrt{2}$ or $2i$

Explanation and steps

Apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = 3\sqrt{2}$ or $2i$.

Question 16 - Q4(iii)

Solve quadratic equation, $x^2 - (5 - i)x + (18 + i) = 0$

Options

A. $x = 3 - 4i$ or $2 + 3i$

B. $x = 3 + 4i$ or $2 - 3i$

C. $x = 5$ or $-i$

D. $x = 3 - 2i$ or $2 + i$

Correct answer: $x = 3 - 4i$ or $2 + 3i$

Explanation and steps

Apply the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = 3 - 4i$ or $2 + 3i$.

Question 17 - Q4(iv)

Solve quadratic equation, $(2 + i)x^2 - (5 - i)x + 2(1 - i) = 0$

Options

A. $x = 1 - i$ or $(4 - 2i)/(5)$

B. $x = 1 + i$ or $(4 + 2i)/(5)$

C. $x = 1$ or $(1 - 2i)/(5)$

D. $x = -1 - i$ or $(2 - 4i)/(5)$

Correct answer: $x = 1 - i$ or $(4 - 2i)/(5)$

Explanation and steps

Apply the quadratic formula $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$, simplify negative radicals using $i = \sqrt{-1}$, and reduce. The solution is $x = 1 - i$ or $(4 - 2i)/(5)$.